

Security risk assessment report

Part 1: Select up to three hardening tools and methods to implement

Multi-factor authentication (MFA): To add a layer of verification that does not rely solely on the password.

Firewall maintenance (including port filtering): To establish critical rules that control traffic flow.

Password policies (Security configuration): To eliminate the use of default keys and prevent employees from sharing their credentials.

Part 2: Explain your recommendations

Recommendation 1: Multi-Factor Authentication (MFA) Why is it effective?: MFA is the most robust defense against credential theft or exchange. Even if an attacker guesses the administrator's password or uses a password shared by an employee, they will not be able to access the system without the second factor (such as a code in an app or a text message). This directly neutralizes three of the four vulnerabilities found.

How often should it be applied?: It should always be required, at every login attempt, especially for accessing databases and internal resources of the organization.

Recommendation 2: Firewall Maintenance and Configuration Why is it effective?: Firewall maintenance involves reviewing and updating configurations to implement port filtering. This allows blocking unwanted communications and ensuring that only legitimate traffic enters or leaves the network. By establishing clear rules, you prevent threat agents from entering through unnecessarily exposed services.

How often should it be applied?: Regular review (monthly or bi-weekly) is recommended, and after any significant change in the network infrastructure to ensure that the rules remain effective against new threats.